10.7 Graphs of circles and exponentials

We know the circle as a common shape in geometry. We can also describe a circle using a rule and as a graph on the Cartesian plane.

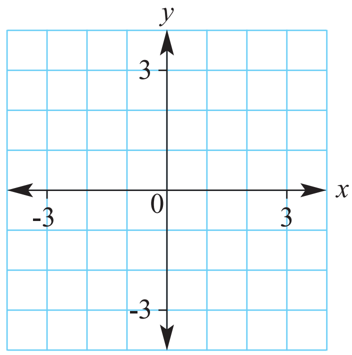
We can also use graphs to illustrate exponential relationships. The population of the world, for example, or the balance of an investment account, can be described using exponential rules that include indices. The rule $A = 100\,000(1.05)^t$ describes the account balance of \$100 000 invested at 5% p.a. compound interest for t years.

Let's start: Plotting non-linear curves

A graph has the rule $x^2 + y^2 = 9$.

- If $x = 0$ what are the two values of y?
- If $x = 1$ what are the two values of y?
- If $x = 4$ are there any values of y? Discuss.
- Complete this table of values.

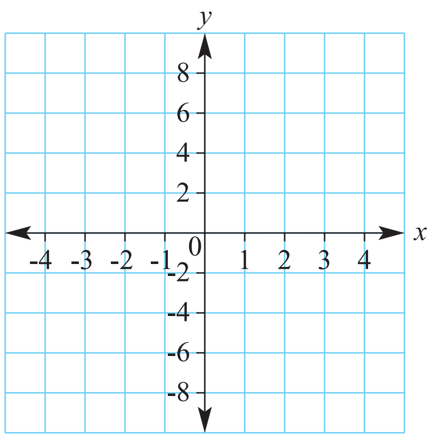
x	-3	-2	-1	0	1	2	3
y		$\pm\sqrt{5}$					



- Now plot all your points on a number plane and join them to form a smooth curve.
- What shape have you drawn and what are its features?
- How does the radius of your circle relate to the equation?

Complete this table and graph the rule $y = 2^x$ before discussing the points below.

x	-3	-2	-1	0	1	2	3
$y = 2^x$	$\frac{1}{8}$			1		4	



- Discuss the shape of the graph.
- Where does the graph cut the y-axis?
- Does the graph have an x-intercept? Why not?

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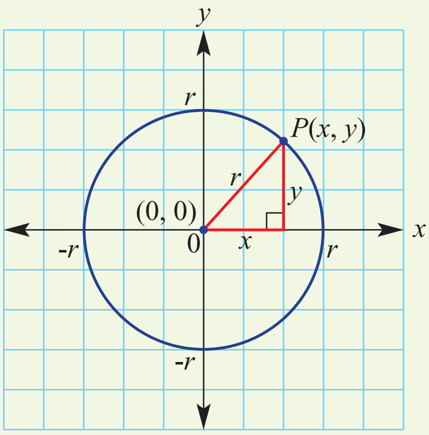
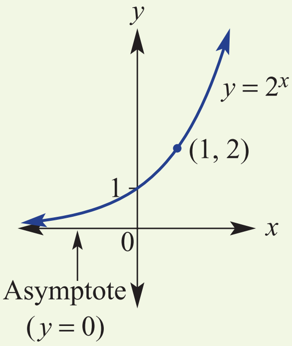
Key ideas

- The Cartesian equation of a circle with centre (0, 0) and radius r is given by $x^2 + y^2 = r^2$.



Using Pythagoras' theorem
$a^2 + b^2 = c^2$ gives $x^2 + y^2 = r^2$

- An **asymptote** is a line that a curve approaches but never touches. The curve gets closer and closer to the line so that the distance between the curve and the line approaches zero, but the curve never meets the line so the distance is never zero.



- A simple **exponential** rule is of the form $y = a^x$, where $a > 0$ and $a \neq 1$.
  - y-intercept is (0, 1).
  - $y = 0$ is the equation of the asymptote.

Asymptote A line whose distance to a curve approaches zero, and the curve never touches it

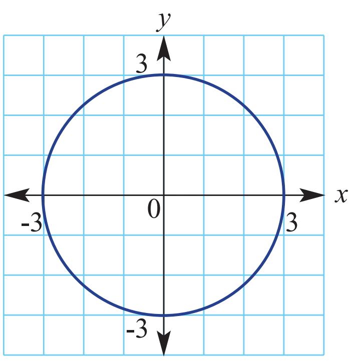
Exponential (notation) A way of representing repeated multiplication of the same number

Exercise 10G

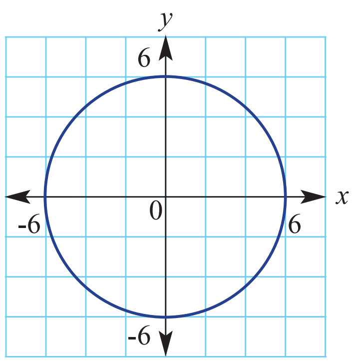
Understanding

1 Write the coordinates of the centre and give the radius of these circles.

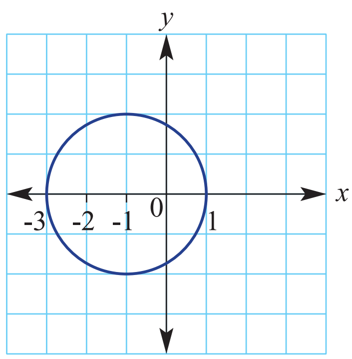
a



b



c



2 Solve these equations for the unknown variable, correct to one decimal place where necessary.

a $x^2 + 2^2 = 9$ **b** $x^2 + 3^2 = 25$ **c** $5^2 + y^2 = 36$

There are two answers for each. ±4 means the answers are +4 and -4.

3 A circle has equation $x^2 + y^2 = r^2$. Complete these sentences.

a The centre of the circle is _______.

b The radius of the circle is _______.

4 Evaluate:

a 2^0 **b** 2^1 **c** 2^4 **d** 3^0 **e** 3^1

f 3^3 **g** 4^0 **h** 4^2 **i** 5^0 **j** 5^2

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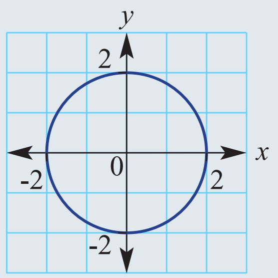
Fluency

Example 20 Sketching a circle

Complete the following for the equation $x^2 + y^2 = 4$.

a State the coordinates of the centre.
b State the radius.
c Find the values of y when $x = 1$, correct to one decimal places.
d Find the values of x when $y = 0$.
e Sketch a graph showing intercepts.

Solution	Explanation
a $(0, 0)$	$(0, 0)$ is the centre for all circles $x^2 + y^2 = r^2$.
b $r = 2$	$x^2 + y^2 = r^2$ so $r^2 = 4$.
c $x^2 + y^2 = 4$ $1^2 + y^2 = 4$ $y^2 = 3$ $y = \pm 1.7$	Substitute $x = 1$ and solve for y. $y^2 = 3$, so $y = \pm\sqrt{3}$ $\sqrt{3} \approx 1.7$
d $x^2 + 0^2 = 4$ $x^2 = 4$ $x = \pm 2$	Substitute $y = 0$. Solve for x. Both $(-2)^2$ and $2^2 = 4$.
e 	Draw a circle with centre (0, 0) and radius 2. Label intercepts.

5 A circle has equation $x^2 + y^2 = 9$. Complete the following.

a State the coordinates of the centre.
b State the radius.
c Find the values of y when $x = 2$, correct to one decimal place.
d Find the values of x when $y = 0$.
e Sketch a graph showing intercepts.

If $x^2 + y^2 = r^2$ then r is the radius.

If $y^2 = 5$, then $y = \pm\sqrt{5}$.

6 Complete the following for the equation $x^2 + y^2 = 25$.

a State the coordinates of the centre.
b State the radius.
c Find the values of y when $x = 4$.
d Find the values of x when $y = 0$.
e Sketch a graph showing intercepts.

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